MTH-ALG, Algebra Problem Sets Solutions

Atonu Roy Chowdhury, achowdhu@ictp.it

Contents

1 Problem Set 1 2

2 Problem Set 2 11

3 Problems on Groups from Lothar Göttsche's Notes 22

§1 Problem Set 1

Exercise 1.1. Let

$$G = \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/20\mathbb{Z}. \tag{1.1}$$

(a) How many elements of G have order 60?

(b) How many cyclic subgroups of G have order 30?

(c) How many injective homomorphisms

$$f : \mathbb{Z}/12\mathbb{Z} \longrightarrow G \tag{1.2}$$

are there?

Solution. (a) We shall prove something first:

> **Claim 1.1:** If $a \in G$ has order n, and $b \in H$ has order m, then $\operatorname{ord}((a,b)) = \operatorname{lcm}(n,m)$ in $G \times H$.
>
> *Proof.* Suppose $(a,b)^k = (e,e)$. Then $a^k = b^k = e$. This gives us that $n \mid k$ and $m \mid k$. Hence,
>
> $$\operatorname{lcm}(n,m) \mid k. \tag{1.3}$$
>
> Furthermore,
>
> $$(a,b)^{\operatorname{lcm}(n,m)} = \left(a^{\operatorname{lcm}(n,m)}, b^{\operatorname{lcm}(n,m)}\right) = (e,e)\,. \tag{1.4}$$
>
> Hence, $\operatorname{ord}((a,b)) = \operatorname{lcm}(n,m)$. □

Now, (a,b) will have order 60 if $\operatorname{lcm}(\operatorname{ord}(a), \operatorname{ord}(b)) = 60$. Since $\operatorname{ord}(a) \in \{1,2,3,6\}$ and $\operatorname{ord}(b) \in \{1,2,4,5,10,20\}$, $\operatorname{lcm}(\operatorname{ord}(a), \operatorname{ord}(b)) = 60$ can only happen in the following cases:

- $\operatorname{ord}(a) = 3, \operatorname{ord}(b) = 20$. Then the choices for b are all the numbers coprime to 20, so we have $\varphi(20) = 8$ choices. The choices for a are $2, 4$. Hence there are $8 \times 2 = 16$ possibilities.
- $\operatorname{ord}(a) = 3, \operatorname{ord}(b) = 20$. Then the choices for b are all the numbers coprime to 20, so we have $\varphi(20) = 8$ choices. The choices for a are $1, 5$. Hence there are $8 \times 2 = 16$ possibilities.

Hence, there are $16 + 16 = 32$ elements of order 60.

(b) Let's first calculate how many elements (a,b) are there of order 30. As before, the possibilities are:

- $\operatorname{ord}(a) = 3, \operatorname{ord}(b) = 10$. Then the choices for b are multiples of 2 by all the numbers coprime to 10, so we have $\varphi(10) = 4$ choices. The choices for a are $2, 4$. Hence there are $4 \times 2 = 8$ possibilities.

- $\bullet \operatorname{ord}(a) = 6, \operatorname{ord}(b) = 10$. Then the choices for b are multiples of 2 by all the numbers coprime to 10, so we have $\varphi(10) = 4$ choices. There are $\varphi(6) = 2$ choices for a. Hence there are $4 \times 2 = 16$ possibilities.

- $\bullet \operatorname{ord}(a) = 6, \operatorname{ord}(b) = 5$. Then the choices for b are multiples of 4 by all the numbers coprime to 5, so we have $\varphi(5) = 4$ choices. There are $\varphi(6) = 2$ choices for a. Hence there are $4 \times 2 = 16$ possibilities.

Hence, there are $8 + 8 + 8 = 24$ numbers of order 30. A subgroup of order 30 consists of $\varphi(30) = 8$ generators. Hence, the number of subgroups of order 30 is $24/8 = 3$.

(c) Note that f is completely fixed by $f\left(\overline{1}\right)$. For f to be injective, $f\left(\overline{1}\right)$ has to be an element of order 12. Let's now calculate how many elements (a, b) are there of order 12. The possibilities are:

- $\bullet \operatorname{ord}(a) = 3, \operatorname{ord}(b) = 4$. Then the choices for b are multiples of 5 by all the numbers coprime to 4, so we have $\varphi(4) = 2$ choices. The choices for a are $2, 4$. Hence there are $2 \times 2 = 4$ possibilities.

- $\bullet \operatorname{ord}(a) = 6, \operatorname{ord}(b) = 4$. Then the choices for b are multiples of 5 by all the numbers coprime to 4, so we have $\varphi(4) = 2$ choices. There are $\varphi(6) = 2$ choices for a. Hence there are $2 \times 2 = 4$ possibilities.

Hence, there are $4 + 4 = 8$ elements of order 12. As a result, there are 12 injective homomorphisms $f : \mathbb{Z}_{12} \to G$.

■

Exercise 1.2. Let $(G_i)_{i=1}^n$ be a family of normal subgroups of G such that

$$G = \left\langle \bigcup_{i=1}^{n} G_i \right\rangle. \tag{1.5}$$

Show that G is isomorphic to the direct product of the G_i if and only if one of the following conditions holds:

1. For every i,
$$G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle = \{e\}. \tag{1.6}$$

2. Every element $g \in G$ can be written uniquely as a product
$$g = g_1 g_2 \cdots g_n, \qquad g_i \in G_i. \tag{1.7}$$

Solution. We will show the equivalence of the following 3 statements:

1. The map $f : G_1 \times \cdots \times G_n \to G$ defined by

$$f(g_1, g_2, \ldots, g_n) = g_1 g_2 \cdots g_n \tag{1.8}$$

is an isomorphism.

2. Every element $g \in G$ can be written uniquely as a product

$$g = g_1 g_2 \cdots g_n, \qquad g_i \in G_i. \tag{1.9}$$

3. For every i,

$$G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle = \{e\}. \tag{1.10}$$

Claim 1.2: Assuming 2, if $g_i \in G_i$ and $g_j \in G_j$ for $i \neq j$, then $g_i g_j = g_j g_i$.

Proof. WLOG, assume $i < j$. Since G_i is normal, $g_j g_i g_j^{-1} \in G_i$, so let's call this element g_i'. Therefore,

$$g_j g_i = g_i' g_j. \tag{1.11}$$

Similarly, since G_j is normal, $g_i^{-1} g_j g_i \in G_j$, so let's call this element g_j'. Therefore,

$$g_j g_i = g_i g_j'. \tag{1.12}$$

But this gives two different representations of the element $g_j g_i$:

$$g_j g_i = e \cdots e \underbrace{g_i'}_{i\text{-th slot}} e \cdots e \underbrace{g_j}_{j\text{-th slot}} e \cdots e = e \cdots e \underbrace{g_i}_{i\text{-th slot}} e \cdots e \underbrace{g_j'}_{j\text{-th slot}} e \cdots e. \tag{1.13}$$

By the uniqueness, we have $g_i = g_i'$ and $g_j = g_j'$, proving that $g_j g_i = g_i g_j$. □

Claim 1.3: Assuming 3, if $g_i \in G_i$ and $g_j \in G_j$ for $i \neq j$, then $g_i g_j = g_j g_i$.

Proof. Consider the element $x := g_i g_j g_i^{-1} g_j^{-1}$. By the normality of G_j, $g_i g_j g_i^{-1} \in G_j$, so that $x \in G_j$. Similarly, by the normality of G_i, $g_j g_i^{-1} g_j^{-1} \in G_i$, so that $x \in G_i$. Since the intersection $G_i \cap G_j = \{e\}$, we can conclude that $x = e$, i.e. $g_i g_j = g_j g_i$. □

(1⇒2) Trivial.

(2⇒3) Let $x \in G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle$. Since $x \in G_i$, it can be uniquely represented as

$$x = e \cdots e \underbrace{x}_{i\text{-th slot}} e \cdots e. \tag{1.14}$$

Since $x \in \left\langle \bigcup_{j\neq i} G_j \right\rangle$, x can be written as a string of elements of G_j. Since elements from different G_j's commute, we can group the elements from the same G_j's and then form

$$x = g_1 g_2 \cdots g_{i-1} e g_{i+1} \cdots g_n. \tag{1.15}$$

by uniqueness, we must have that all g_j's must be e, and also $x = e$. Hence, $G_i \cap \left\langle \bigcup_{j\neq i} G_j \right\rangle = \{e\}$.

(3⇒1) First, we'll show that f is a homomorphism.

$$f(g_1, \ldots, g_n) f\left(g_1', \ldots, g_n'\right) = (g_1 \cdots g_n)\left(g_1' \cdots g_n'\right). \tag{1.16}$$

Since elements from different G_i's commute, this can be rewritten as

$$(g_1 \cdots g_n)\left(g_1' \cdots g_n'\right) = \left(g_1 g_1'\right)\left(g_2 g_2'\right) \cdots \left(g_n g n'\right) = f\left(g_1 g_1', \ldots, g_n g_n'\right), \tag{1.17}$$

i.e. f is a homomorphism. f is surjective indeed: since $G = \langle \bigcup_{i=1}^n G_i \rangle$, every element $g \in G$ can be written as a string of elements of G_i's. Since elements from different G_i's commute, we can group the elements from the same G_i's and then form

$$g = g_1 g_2 \cdots g_n, \quad \text{where } g_i \in G_i \text{ for each } i. \tag{1.18}$$

Now, suppose $(g_1, \ldots, g_n) \in \operatorname{Ker} f$. Then

$$e = g_1 g_2 \cdots g_n. \tag{1.19}$$

From this, we can write

$$\begin{aligned} g_i^{-1} &= g_i^{-1}\ g_1 g_2 \cdots g_n = g_1 g_2 \cdots g_{i-1}\left(g_i^{-1} g_i\right) g_{i+1} \cdots g_n \\ &= g_1 g_2 \cdots g_{i-1} g_{i+1} \cdots g_n \in \left\langle \bigcup_{j\neq i} G_j \right\rangle. \end{aligned} \tag{1.20}$$

(Here the reordering is possible because g_i^{-1} commutes with the other g_j's.) Since $G_i \cap \left\langle \bigcup_{j\neq i} G_j \right\rangle = \{e\}$, this forces that $g_i = e$. This is true for each i. Hence, $\operatorname{Ker} f$ is trivial, and hence it's injective. This completes the proof. ■

Exercise 1.3. Let G be a group, and define

$$\Delta = \{(x, x) \mid x \in G\}. \tag{1.21}$$

(a) Show that Δ is a subgroup of $G \times G$.

(b) Prove that Δ is a normal subgroup of $G \times G$ if and only if G is abelian. Moreover, in this case, show that the quotient group $(G \times G)/\Delta$ is isomorphic to G.

Solution. (a) Take $(x,x),(y,y) \in \Delta$. Then

$$(x,x) * (y,y)^{-1} = (x,x) * \left(y^{-1}, y^{-1}\right) = \left(xy^{-1}, xy^{-1}\right) \in \Delta. \tag{1.22}$$

Therefore, Δ is a subgroup of $G \times G$.

(b) All subgroups of an abelian group are normal. So if G is abelian, so is $G \times G$, and hence Δ is a normal subgroup.

Suppose Δ is a normal subgroup. Then given any $(g,h) \in G \times G$,

$$(g,h) * (x,x) * (g,h)^{-1} = (g,h) * (x,x) * \left(g^{-1}, h^{-1}\right) = \left(gxg^{-1}, hxh^{-1}\right). \tag{1.23}$$

This is in Δ, by hypothesis that Δ is normal. Therefore,

$$gxg^{-1} = hxh^{-1} \tag{1.24}$$

Since this holds for every $g, x, h \in G$, we choose $h = e$, so we have

$$gxg^{-1} = x \implies gx = xg, \tag{1.25}$$

proving that G is abelian. Suppose G is abelian, then we define

$$\begin{aligned} f : G \times G &\to G \\ (x,y) &\mapsto xy^{-1}. \end{aligned} \tag{1.26}$$

This is a homomorphism, because

$$f\left((g,h) * (x,y)\right) = f\left(gx, hy\right) = (gx)\left(hy\right)^{-1} = gxy^{-1}h^{-1} \tag{1.27}$$

$$= \left(gh^{-1}\right)\left(xy^{-1}\right) = f\left(g,h\right) f\left(x,y\right). \tag{1.28}$$

Therefore, f is a homomorphism. Now,

$$(x,y) \in \operatorname{Ker} f \iff xy^{-1} = e \iff x = y. \tag{1.29}$$

Therefore, $\operatorname{Ker} f = \Delta$. f is surjective, since any $g \in G$ is $f(g,e) = g$. Hence, by the first isomorphism theorem,

$$\operatorname{im} f \cong \frac{\operatorname{dom} f}{\operatorname{Ker} f} \implies G \cong \frac{G \times G}{\Delta}. \tag{1.30}$$

■

Exercise 1.4. A subgroup M of a group G is called *maximal* if $M \neq G$ and for every subgroup L of G,

$$M \subsetneq L \subseteq G \implies L = G. \tag{1.31}$$

(a) Let K, M be subgroups of a group G such that $K \trianglelefteq G$ and $K \subseteq M$. Prove that M is a maximal subgroup of G if and only if M/K is a maximal subgroup of G/K.

(b) Prove that in a finite group G, every proper subgroup $H \neq G$ is contained in some maximal subgroup of G.

(c) Prove that in a finite abelian group different from the trivial group, the maximal subgroups are precisely those of prime index.

(d) Is the previous point true if we remove the assumption that the group is abelian?

Solution. (a) Let $\pi : G \to G/K$ denote the projection map. Suppose M is a maximal subgroup of G. Assume for the sake of contradiction that M/K is **NOT** maximal. Then there exists another subgroup

$$\frac{M}{K} \subsetneq X \subsetneq \frac{G}{K}. \tag{1.32}$$

Now consider $\pi^{-1}(X)$.

Claim 1.4: If $f : G_1 \to G_2$ is a homomorphism, and $H_2 \subseteq G_2$ is a subgroup, then $f^{-1}(H_2) \subseteq G_1$ is a subgroup.

Proof. Take $x, y \in f^{-1}(H_2)$. Then $f(x), f(y) \in H_2$. Since $H_2 \subseteq G_2$ is a subgroup,

$$f\left(xy^{-1}\right) = f(x) f(y)^{-1} \in H_2, \tag{1.33}$$

proving that $xy^{-1} \in f^{-1}(H_2)$. So $f^{-1}(H_2) \subseteq G_1$ is a subgroup. □

Now, $\pi^{-1}(X)$ is a subgroup of G, that contains $\pi^{-1}(M/K) = M$. Hence, either $\pi^{-1}(X) = M$ or $\pi^{-1}(X) = G$.

$$\pi^{-1}(X) = \{g \in G \mid gK \in X\}. \tag{1.34}$$

If $\pi^{-1}(X) = G$, then $X = G/K$; and if $\pi^{-1}(X) = M$, then $X = M/K$; both of which contradict (1.32).

Conversely, suppose M/K is a maximal subgroup in G/K. Assume for the sake of contradiction that M is **NOT** maximal. Then there exists another subgroup

$$M \subsetneq H \subsetneq G. \tag{1.35}$$

Since image of a subgroup is also a subgroup, we have $\frac{M}{K} \subset \frac{H}{K} \subset \frac{G}{K}$. By the maximality of M/K, either $H/K = M/K$ or $H/K = G/K$.

Claim 1.5: For any subgroup $L \supseteq K$, $\pi^{-1}(L/K) = L$.

Proof. It's clear that $L \subseteq \pi^{-1}(L/K)$. Now take any $g \in \pi^{-1}(L/K)$. Then $gK \in L/K$.

$$gK = lK \text{ for some } l \in L. \tag{1.36}$$

Therefore, $l^{-1}g \in K \subseteq L$. Hence, $g = l\left(l^{-1}g\right) \in L$. As a result, $L \supseteq \pi^{-1}(L/K)$. □

If $H/K = M/K$, by taking π^{-1}, we get $H = K$; and if $H/K = G/K$, then $H = G$; both of which contradict (1.35).

(b) Let H be a proper subgroup of a finite group G. We construct a sequence of subgroups

$$H =: H_0 \subsetneq H_1 \subsetneq H_2 \subsetneq \cdots \tag{1.37}$$

If H_i is maximal, then we are done, since H is contained in a maximal subgroup! Otherwise, there exists some $g_{i+1} \in G - H_i$ such that

$$H_i \subsetneq H_{i+1} := \langle H_i, g_{i+1} \rangle \subsetneq G. \tag{1.38}$$

If there is no such g_{i+1}, then H_i is indeed maximal, because adding one element to H_i and making a group with it makes it the whole G.

Now, since G is a finite group, this sequence can't go on forever. Therefore, it has to stop somewhere, and thus we find a maximal subgroup that contains H.

(c) Let M be a subgroup of prime index, i.e. $[G : M] = p$. Let's pick a subgroup L with

$$M \subseteq L \subseteq G. \tag{1.39}$$

Then

$$[G : M] = [G : L]\,[L : M]\,. \tag{1.40}$$

(This follows from the fact that for finite groups $[G : H] = \frac{|G|}{|H|}$.) Since $[G : M]$ is a prime, it forces either $[G : L] = 1$, or $[L : M] = 1$; consequently $G = L$ or $L = M$. Therefore, M is maximal.

Now suppose M is a maximal subgroup in a finite abelian group G. Since G is abelian, M is normal, so we can form the quotient group G/M. By part (a), since M is maximal, we can conclude that $\{\overline{e}\} = M/M$ is a maximal subgroup of G/M.

Claim 1.6: If $\{e\}$ is a maximal subgroup of $G \neq \{e\}$, then $G \cong \mathbb{Z}_p$ for some prime p.

Proof. Adding one single non-identity element makes it the whole group, so for every $a \in G$ with $a \neq e$,

$$\langle a \rangle = G. \tag{1.41}$$

Hence, G is cyclic. If $|G|$ was infinite, then it would have subgroups other than the trivial one. But as $\{e\}$ is a maximal subgroup, it's the only subgroup. Therefore, $|G|$ can't be infinite.

Now, if $|G|$ was composite, say $|G| = mn$, then $a^{mn} = e$, proving a^m has order n. This gives rise to a subgroup of size n, namely $\langle a^m \rangle$, which is a contradiction! Therefore, $|G|$ is prime. □

Now, $\{\overline{e}\} = M/M$ is a maximal subgroup of G/M, proves that $|G/M| = p$. Therefore,

$$M \subsetneq G \text{ is maximal } \iff [G : M] \text{ is prime.} \tag{1.42}$$

(d) No. We can consider the group $G = A_4$ of alternating group of degree 4, which contains all even permutations of $\{1, 2, 3, 4\}$. The subgroup

$$H = \langle (1\ 2\ 3) \rangle \tag{1.43}$$

has size 3, so that $[G : H] = 2^2$ is not a prime. However, H is maximal. Indeed, if there exists another subgroup

$$H \subsetneq K \subsetneq G, \tag{1.44}$$

then $|K|$ divides $|G| = 12$, and $|H| = 3$ divides $|K|$, forces $|K| = 6$. If K contains a product of two 2-cycles, e.g. $(1\ 2)\,(3\ 4)$, then

$$(1\ 2\ 3)\,(1\ 2)\,(3\ 4) = (1\ 3\ 4) \in K \tag{1.45}$$

Since K is a group, it contains the inverse $(1\ 4\ 3)$ as well. These are all the 6 elements of K. But then

$$(1\ 2)\,(3\ 4)\,(1\ 2\ 3) = (2\ 4\ 3) \tag{1.46}$$

that gives one more element in K, which is a contradiction!

Therefore, K doesn't contain any product of two 2-cycles, i.e. the only non-identity elements in K are 3-cycles. But every 3-cycle comes with its inverse as well, and no 3-cycle is its own inverse. Therefore, $|K|$ can't be even! Thus we arrive at a contradiction. Hence, H is maximal.

■

Exercise 1.5. Let p be a prime number.

(a) Determine all elements of order dividing p in S_n. Denote by a_n the number of such elements and derive a formula for a_n.

(b) Prove the identity

$$\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n = e^{x + \frac{x^p}{p}}. \tag{1.47}$$

Solution. (a) Every element $\sigma \in S_n$ can be decomposed into disjoint cycles. The disjoint cycles commute. Now, if $\operatorname{ord} \sigma \mid p$, then we must have $\sigma^p = e$. This will happen if and only if $c_i^p = e$ for each disjoint cycle c_i in σ.

Now, for a cycle c_i, $c_i^p = e$ happens if the length of the cycle is either 1 or p. Therefore, a_n will be the number of ways we can decompose $\{1, 2, \ldots, n\}$ into cycles of length 1 or p.

Now, suppose $\sigma \in S_n$ consists of m disjoint cycles of length p, and the other $n - mp$ elements are fixed points (i.e. cycles of length 1). Here, m ranges from 0 to $\left\lfloor \frac{n}{p} \right\rfloor$. Therefore,

$$a_n = \sum_{m=0}^{\left\lfloor \frac{n}{p} \right\rfloor} \text{number of } \sigma\text{'s with } m \text{ disjoint cycles of length } p. \tag{1.48}$$

First, we can choose the numbers involved in the p-cycles in $\binom{n}{mp}$ ways. Now, we can partition them into m sets of size p, which can be done in

$$\binom{mp}{p} \cdot \binom{mp-p}{p} \cdot \cdots \cdot \binom{p}{p} = \frac{(mp)!}{(p!)^m} \tag{1.49}$$

many ways. Now, each cycles can be permuted in $(p-1)!$ different ways. Finally, the cycles themselves can permute in $m!$ ways among themselves. So we need to divide by that. Therefore, the number of σ's with m disjoint cycles of length p is

$$\binom{n}{mp} \cdot \frac{(mp)!}{(p!)^m} \cdot ((p-1)!)^m \cdot \frac{1}{m!} = \frac{n!}{p^m m! (n-mp)!}. \tag{1.50}$$

Hence,

$$a_n = \sum_{m=0}^{\left\lfloor \frac{n}{p} \right\rfloor} \frac{n!}{p^m m! (n-mp)!}. \tag{1.51}$$

(b) Using (1.51), we have

$$f(x) := \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n = \sum_{n=0}^{\infty} \sum_{m=0}^{\left\lfloor \frac{n}{p} \right\rfloor} \frac{x^n}{p^m m! (n-mp)!}. \tag{1.52}$$

We can swap the sum into

$$f(x) = \sum_{m=0}^{\infty} \sum_{n=mp}^{\infty} \frac{x^n}{p^m m! (n-mp)!}. \tag{1.53}$$

Suppose $k = n - mp$, so that $n = k + mp$. Then the sum can be rewritten as

$$f(x) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{x^{k+mp}}{p^m m! k!} = \sum_{m=0}^{\infty} \frac{1}{m!} \left(\frac{x^p}{p} \right)^m \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^{\frac{x^p}{p}} e^x = e^{\frac{x^p}{p} + x}. \tag{1.54}$$

■

§2 Problem Set 2

Exercise 2.1. Find all p-Sylow subgroups of the following groups:

(a) $\mathbb{Z}/n\mathbb{Z}$;

(b) $\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}$;

(c) the dihedral group D_n (the group of symmetries of a regular n-polygon).

Solution. Before proceeding, we claim that for abelian groups, the p-Sylow subgroup is unique! Indeed, if H_1 and H_2 are two p-Sylow subgroups of G, then by Sylow's Second Theorem, they are related by a conjugation, i.e.

$$H_2 = gH_1g^{-1} \text{ for some } g \in G. \tag{2.1}$$

But if G is abelian, we have

$$H_2 = gH_1g^{-1} = H_1. \tag{2.2}$$

Furthermore, notice that if G is not necessarily abelian, but a p-Sylow subgroup happens to be normal, then also the p-Sylow subgroup is unique. Because if H_1 is a p-Sylow subgroup which happens to be normal, and H_2 is another p-Sylow subgroup, then

$$H_2 = gH_1g^{-1} = H_1. \tag{2.3}$$

We shall use these two properties below.

(a) Let $n = p^k m$, where $p \nmid m$. Since the group is abelian, there is exactly one p-Sylow subgroup, i.e. subgroup of order p^m. That's precisely the subgroup generated by m, since it has order p^k, i.e.

$$H = \langle m \rangle = \left\{0, m, 2m, 3m, \ldots, \left(p^k - 1\right) m\right\}. \tag{2.4}$$

(b) $|G| = 6 \cdot 12 = 2^3 \cdot 3^2$. Since the group is abelian, there is exactly one p-Sylow subgroup. For $p = 2$, we can choose a copy of $\mathbb{Z}_2$ inside $\mathbb{Z}_6$, and a copy of $\mathbb{Z}_4$ inside $\mathbb{Z}_{12}$, and take their direct product, and that will be the subgroup of order 2^3.

$$H = \{0, 3\} \times \{0, 3, 6, 9\} \subset \mathbb{Z}_6 \times \mathbb{Z}_{12}. \tag{2.5}$$

For $p = 3$, we can choose a copy of $\mathbb{Z}_3$ inside $\mathbb{Z}_6$, and a copy of $\mathbb{Z}_3$ inside $\mathbb{Z}_{12}$, and take their direct product, and that will be the subgroup of order 3^2.

$$H' = \{0, 2, 4\} \times \{0, 4, 8\} \subset \mathbb{Z}_6 \times \mathbb{Z}_{12}. \tag{2.6}$$

(c) $|D_n| = 2n$. Suppose $p \neq 2$. Then $p \mid 2n$ if and only if $p \mid n$. We write

$$2n = 2 \cdot p^k \cdot m, \tag{2.7}$$

where $p \nmid m$. Let a be rotation by $\frac{2\pi}{n}$. Then $\langle a \rangle$ is an order-n subgroup of D_n. We take

$$H = \langle a^m \rangle = \left\{ e, a^m, a^{2m}, \ldots, a^{\left(p^k-1\right)m} \right\}, \tag{2.8}$$

which is a subgroup of order p^k, i.e. a p-Sylow subgroup. Now, we claim that H is normal in D_n. Indeed, all the rotations commute with all the elements of H. If b is a reflection, then we have

$$ba^ib^{-1} = a^{-i}, \tag{2.9}$$

(this can be proven by induction on i), and hence

$$\left(a^jb\right)\ a^i\ \left(a^jb\right)^{-1} = a^jba^iba^{-j} = a^ja^{-i}a^{-j} = a^{-i}. \tag{2.10}$$

Therefore, H is normal, and hence there is only one p-Sylow subgroup for $p \neq 2$.

Now suppose $p = 2$. If $2 \nmid n$, then

$$n_2 \mid n \implies n_2 \leq n. \tag{2.11}$$

We can then exhibit n subgroups of order 2, namely the ones containing one reflection and the identity:

$$H_i = \left\{ e, a^ib \right\}, \quad \text{for } i = 0, 1, 2, \ldots, n-1. \tag{2.12}$$

Now consider the case $2 \mid n$. Suppose $n = 2^km$, where m is odd. Then

$$|D_n| = 2n = 2^{k+1}m. \tag{2.13}$$

Then

$$n_2 \mid m \implies n_2 \leq m. \tag{2.14}$$

We now exhibit m subgroups of order 2^{k+1}. These will be isomorphic to D_{2^k}. We define

$$H_i = \left\langle a^m, a^ib \right\rangle, \quad \text{for } i = 0, 1, 2, \ldots, m-1 \tag{2.15}$$

Here, $a' := a^m$ is a rotation of order 2^k (since $a^{2^km} = a^n = e$), and $b' := a^ib$ is a reflection of degree 2, and

$$b'a'b' = \left(a^ib\right)\left(a^m\right)\left(a^ib\right) = a^i\left(ba^{i+m}b\right) = a^ia^{-i-m} = a^{-m} = \left(a'\right)^{-1}, \tag{2.16}$$

and therefore,

$$H_i = \left\langle a', b' \mid \left(a'\right)^{2^k} = \left(b'\right)^2 = e,\ b'a'b' = \left(a'\right)^{-1} \right\rangle \cong D_{2^k}, \tag{2.17}$$

so each H_i is a subgroup of size 2^{k+1}, i.e. a 2-Sylow subgroup. Since $n_2 \leq m$, and we have exhibited m subgroups, these are the all of them!

■

Exercise 2.2. Find all 2-Sylow subgroups of the following groups:

(a) the symmetric group S_4;

(b) the symmetric group S_5.

Solution. (a) $|S_4| = 24 = 2^3 \cdot 3$. So we need to find all subgroups of size 8. By Sylow's theorems, if n_2 is the number of 2-Sylow subgroups,

$$n_2 \mid 3 \implies n_2 \in \{1, 3\}\,. \tag{2.18}$$

We claim that there are 3 of them, by finding 3 subgroups of order 8. First, note that there's a subgroup isomorphic to D_4.

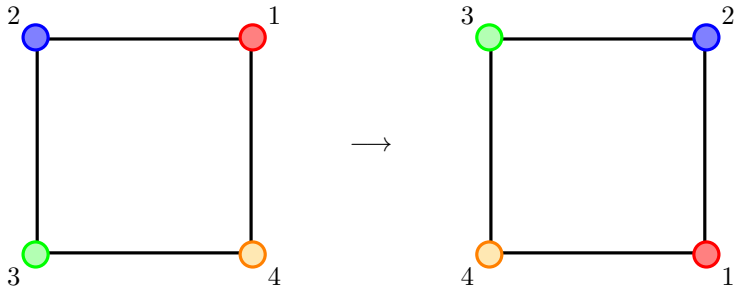


Figure 1: Rotation in D_4, which can be realized as the element (1 2 3 4). To see the resemblence, in the left figure, 1 is in the top-right position, which is occupied by 2 in the right figure, so $\sigma(1) = 2$, etc.

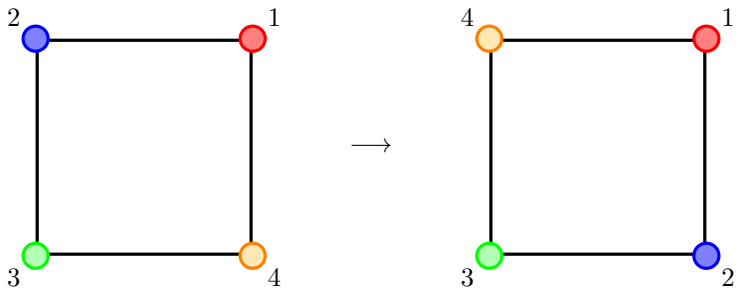


Figure 2: Reflection in D_4, which can be realized as the element (2 4). To see the resemblence, in the left figure, 2 is in the top-left position, which is occupied by 4 in the right figure, so $\sigma(2) = 4$, etc.

These two generate D_4, so we can take the corresponding subgroup in S_4

$$G_1 = \langle (1\ 2\ 3\ 4)\,, (2\ 4) \rangle\,, \tag{2.19}$$

which is isomorphic to D_4, and hence a subgroup of order 8, i.e. a 2-Sylow subgroup. To find the others, it's sufficient to change the labels in the vertices of the square. There are

3-possible ways we can cyclically rename the vertices, which gives the other two subgroups:

$$G_2 = \langle (1\ 3\ 2\ 4), (1\ 2) \rangle, \tag{2.20}$$

$$G_3 = \langle (1\ 2\ 4\ 3), (1\ 4) \rangle. \tag{2.21}$$

These are all the 2-Sylow subgroups of size 8.

(b) $|S_5| = 120 = 2^3 \cdot 15$. So we need to find all subgroups of size 8. By Sylow's theorems, if n_2 is the number of 2-Sylow subgroups,

$$n_2 \mid 15 \implies n_2 \in \{1, 3, 5, 15\}. \tag{2.22}$$

Note that, given any 4-element subset of $\{1, 2, 3, 4, 5\}$, we can find 3 subgroups of size 8, as per part (a). The total number of 4-element subsets of $\{1, 2, 3, 4, 5\}$ is $\binom{5}{4} = 5$. Hence, this already gives us a recipe to build $5 \times 3 = 15$ subgroups of order 8. Since $n_2 \leq 15$, these are all of them!

■

Exercise 2.3. (a) Show that there are no simple groups of order 30, 42 or 105.

(b) More generally, show that there are no simple groups of order pqr, where p, q, r are distinct prime numbers.

Solution. We'll just solve (b). Assume for the sake of contradiction that ther exists a simple group G with $|G| = pqr$. WLOG, $p < q < r$.

Note that, we must have $n_r > 1$, because if $n_r = 1$, then the r-Sylow subgroup would've been normal, contradicting the simplicity of G. Also, $n_r \equiv 1 \pmod{r}$ forces that $n_r > r > q > p$. Now, since

$$n_r \mid pq, \tag{2.23}$$

the only possibility is $n_r = pq$. In other words, there are pq subgroups of order r, which gives

$$(\text{number of elements of order } r) = n_r (r-1) = pq (r-1) = |G| - pq. \tag{2.24}$$

Similarly, we must have $n_q > 1$, because if $n_q = 1$, then the q-Sylow subgroup would've been normal, contradicting the simplicity of G. Also, $n_q \equiv 1 \pmod{q}$ forces that $n_q > q > p$. Therefore,

$$(\text{number of elements of order } q) = n_q (q-1) > pq - p. \tag{2.25}$$

Therefore,

$$(\text{number of elements of order } p) = n_p (p-1) < p. \tag{2.26}$$

But this forces $n_p = 1$, which implies the p-Sylow subgroup is normal, contradicting the simplicity of G. Therefore, there is no simple group of order pqr. ■

Exercise 2.4. Consider the action of a finite group G on a finite set X, with N orbits. For every $g \in G$, consider the set

$$\mathrm{Fix}(g) := \{x \in X \mid g \cdot x = x\}. \tag{2.27}$$

(a) Derive Burnside's formula:

$$N = \frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)| \,. \tag{2.28}$$

(b) Apply the previous formula to solve the following problem: count the numbers of bracelets made of 3 red beads and 7 blue beads. *Hint:* Two bracelets are indistinguishable if one is obtained by the other with a rotation or a reflection.

Solution. (a) Consider the set

$$S = \{(g, x) \mid g \cdot x = x\} \,. \tag{2.29}$$

We can count $|S|$ in two different ways. For every $g \in G$, we count the number of $x \in S$ fixed by g. Then for every $x \in X$, we count the number of $g \in G$ that fixes x. Therefore,

$$|S| = \sum_{g \in G} |\mathrm{Fix}(g)| = \sum_{x \in X} |G_x| \,, \tag{2.30}$$

where G_x is the stabilizer of x. Since X is the disjoint union of G-orbits,

$$\sum_{x \in X} |G_x| = \sum_{\text{orbits } Gx} \sum_{y \in Gx} |G_y| \,. \tag{2.31}$$

Now, for $y \in Gx$, $Gx = Gy$, so their stabilizers have the same size, i.e. $|G_x| = |G_y|$.

> **Claim 2.1:** For $y \in Gx$, $|G_x| = |G_y|$
>
> *Proof.* Since $y \in Gx$, $y = g \cdot x$ for some $g \in G$. If $g_0 \in G_x$ is in the stabilizer of x, then $g_0 \cdot x = x$. Now,
>
> $$\left(gg_0g^{-1}\right) \cdot y = \left(gg_0g^{-1}\right) \cdot (g \cdot x) = \left(gg_0g^{-1}g\right) \cdot x = g \cdot (g_0 \cdot x) = g \cdot x = y. \tag{2.32}$$
>
> Therefore, $gg_0g^{-1} \in G_y$, i.e. $G_y = gG_xg^{-1}$. Since conjugation preserves size, $|G_x| = |G_y|$. □

With this, we rewrite (2.31),

$$\sum_{x \in X} |G_x| = \sum_{\text{orbits } Gx} |Gx| \, |G_x| \tag{2.33}$$

By Orbit-Stabilizer theorem, $|G| = |Gx| \, |G_x|$. Hence,

$$\sum_{x \in X} |G_x| = \sum_{\text{orbits } Gx} |Gx| \, |G_x| = N \, |G| \,, \tag{2.34}$$

where N is the number of orbits. Hence,

$$N = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{G} \sum_{g \in G} |\mathrm{Fix}(g)| \,. \tag{2.35}$$

(b) Let $G = D_{10}$, the dihedral group on a regular 10-gon. Then $|G| = 20$. Let X be the set of all bracelets made of 3 red beads and 7 blue beads (without accounting for indistinguishability). Then

$$|X| = \binom{10}{3} = \frac{10 \cdot 9 \cdot 8}{3!} = 120. \tag{2.36}$$

Consider D_{10}'s action on X. Then the number of indistinguishable bracelets will be the number of orbits in this action. By Burnside's lemma, if N is the number of orbits,

$$N = \frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)| \,. \tag{2.37}$$

For each $g \in G$, let's now compute the size of $\mathrm{Fix}(g)$.

- If $g = e$, $|\mathrm{Fix}(e)| = |X| = 120$.
- If g is a rotation by $\frac{2\pi}{10}k$ for $k = 1, \ldots, 9$, then we claim that $\mathrm{Fix}(g) = \varnothing$.

  If we consider the positions (in modulo 10), then a rotation is just adding a constant with the positions. If r_1, r_2, r_3 are the position of the red beads in a configuration, if g fixes that configuration, then we must have

  $$\{r_1, r_2, r_3\} = \{r_1 + a, r_2 + a, r_3 + a\} \pmod{10}. \tag{2.38}$$

  After adding, we have

  $$r_1 + r_2 + r_3 \equiv r_1 + r_2 + r_3 + 3a \pmod{10}. \tag{2.39}$$

  In other words, $10 \mid 3a$. But since 10 and 3 are coprime, this means $a = 0 \pmod 1 0$, i.e. the rotation is the trivial one. Therefore, for nontrivial rotations, $|\mathrm{Fix}(g)| = 0$.
- If g is reflection in a line ℓ through the midpoints of opposite edges, then there are the same number of beads on both sides of ℓ, and we can't arrange the beads of the same color evenly, because there are an odd number of red beads and an odd number of blue beads. So in this case, $|\mathrm{Fix}(g)| = 0$. There are 5 such reflections.
- If g is reflection in a line ℓ through two opposite vertices, then we can choose one of them to be red, and we can choose the other to be blue. This can be done in $\binom{2}{1} = 2$ ways.

  Then if we color one side of ℓ, the other side becomes fixed. On each side of ℓ, we have 4 points, and we can choose one of them to be red, which can be done in $\binom{4}{1} = 4$ ways. Therefore, $|\mathrm{Fix}(g)| = 2 \times 4 = 8$. There are 5 such reflections.

Now,

$$N = \frac{1}{20}\left(120 + 0 \times 9 + 0 \times 5 + 8 \times 5\right) = \frac{160}{20} = 8. \tag{2.40}$$

■

Exercise 2.5 (Frattini's Lemma)**.** Let $H \trianglelefteq G$ be a normal subgroup of a finite group. Let S be a p-Sylow of H. Show that the conjugates of S in G are in H as well and deduce the following:

$$G = HN_G(S), \tag{2.41}$$

where $N_G(S)$ denotes the normalizer of S in G.

Solution. For every $g \in G$, since $S \leq H$,

$$gSg^{-1} \subseteq gHg^{-1} = H, \tag{2.42}$$

so conjugates of S in G are in H as well! Now we shall prove that any $g \in G$ can be written as $g = hn$ for some $h \in H, n \in N_G(S)$.

Note that since $gSg^{-1} \subseteq H$, gSg^{-1} is also a p-Sylow subgroup of H, because conjugation preserves cardinality. Since both S and gSg^{-1} are p-Sylow subgroups of H, by the Second Sylow Theorem, they are related by a conjugation in H, i.e.

$$gSg^{-1} = hSh^{-1} \tag{2.43}$$

for some $h \in H$. This gives

$$h^{-1}gSg^{-1}h = S. \tag{2.44}$$

Since $N_G(S) = \left\{g \mid gSg^{-1} = S\right\}$, by (2.44), $h^{-1}g \in N_G(S)$.

$$h^{-1}g = n \text{ for some } N_G(S), \text{ for some } h \in H \tag{2.45}$$

$$\implies g = hn \text{ for some } N_G(S), \text{ for some } h \in H. \tag{2.46}$$

Therefore, $G = HN_G(S)$. ■

Exercise 2.6 (Short exact sequences)**.** Given a sequence of morphisms

$$N \xrightarrow{a} G \xrightarrow{b} K \tag{2.47}$$

we say that it is exact in G if

$$\operatorname{Ker} b = \operatorname{im} a. \tag{2.48}$$

A sequence of the form

$$e \longrightarrow N \xrightarrow{a} G \xrightarrow{b} K \longrightarrow e \tag{2.49}$$

which is exact in N, G, K is called a short exact sequence (where we denote with e the group with one element).

(a) Show that in the short exact sequence above a is injective and b surjective.

(b) Show that $a(N) \trianglelefteq G$ and

$$K \simeq G/a(N).$$

(c) Let N and K be groups. Show that you can define a short exact sequence:

$$e \longrightarrow N \xrightarrow{\ i\ } N \times K \xrightarrow{\ p\ } K \longrightarrow e \tag{2.50}$$

Solution. (a) Exactness at N gives

$$\operatorname{Ker} a = \operatorname{im} (e \to N) = \{e\}, \tag{2.51}$$

so a is injective. Exactness at K gives

$$\operatorname{im} b = \operatorname{Ker} (K \to e) = K, \tag{2.52}$$

so b is surjective.

(b) Exactness at G gives $a(N) = \operatorname{im} a = \operatorname{Ker} b$. Kernel is always normal, because for $h \in \operatorname{Ker} b$ and $g \in G$,

$$b\left(ghg^{-1}\right) = b(g)\, b(h)\, b(g)^{-1} = b(g)\, b(g)^{-1} = e, \tag{2.53}$$

so $ghg^{-1} \in \operatorname{Ker} b$, proving $\operatorname{Ker} b$ is normal. Now, since b is surjective,

$$\operatorname{im} b \cong \frac{\operatorname{dom} b}{\operatorname{Ker} b} \implies K \cong \frac{G}{\operatorname{Ker} b} = \frac{G}{a(N)}. \tag{2.54}$$

(c) We define

$$i(n) = (n, e_K) \text{ and } p(n, k) = k. \tag{2.55}$$

Then clearly i is injective and p is surjective. Also,

$$\operatorname{im} i = N \times \{e_K\} = \operatorname{Ker} p. \tag{2.56}$$

Hence, the sequence $e \to N \to N \times K \to K \to e$ is exact.

■

Exercise 2.7 (Semidirect products)**.** Let N and K be groups, and let

$$\varphi : K \longrightarrow \operatorname{Aut}(N) \tag{2.57}$$

be a group homomorphism. For each $k \in K$ and $n \in N$, we denote by $\varphi(k)(n)$ the image of n under the automorphism of N corresponding to k via φ.

(a) Define a binary operation on the set-theoretic product $N \times K$ by

$$(n,k) * \left(n',k'\right) = \left(n\ \varphi(k)\left(n'\right), kk'\right). \tag{2.58}$$

Show that this operation defines a group. We denote this group by $N \rtimes_\varphi K$ and call it the semidirect product of N by K with respect to φ.

(b) Show that there is a short exact sequence

$$e \longrightarrow N \xrightarrow{\ i\ } N \rtimes_\varphi K \xrightarrow{\ p\ } K \longrightarrow e \tag{2.59}$$

where

$$i(n) = (n,e) \qquad \text{and} \qquad p(n,k) = k. \tag{2.60}$$

(c) Show that if

$$\varphi : K \longrightarrow \mathrm{Aut}(N) \tag{2.61}$$

is the trivial homomorphism, then $N \rtimes_\varphi K$ is the usual direct product of groups $N \times K$.

(d) Conversely, let G be a group with subgroups $N, K \leq G$ such that

$$N \trianglelefteq G, \qquad N \cap K = \{e\}, \qquad G = NK. \tag{2.62}$$

Show that G is isomorphic to the semidirect product $N \rtimes_\varphi K$, where

$$\varphi(k)(n) = knk^{-1} \tag{2.63}$$

for $n \in N$ and $k \in K$.

Solution. (a) For simplicity of notations, I'll write $\varphi_k = \varphi(k)$. Since φ is a homomorphism, this will give us

$$\varphi_{kk'} = \varphi\left(kk'\right) = \varphi\left(k\right) \circ \varphi\left(k'\right) = \varphi_k \circ \varphi k'. \tag{2.64}$$

Associativity:

$$\begin{aligned}
\left[(n,k) * \left(n',k'\right)\right] * \left(n'',k''\right) &= \left(n\ \varphi_k\left(n'\right), kk'\right) * \left(n'',k''\right) \\
&= \left(n\ \varphi_k\left(n'\right)\ \ \varphi_{kk'}\left(n''\right), (kk')k''\right) \\
&= \left(n\ \varphi_k\left(n'\right)\ \ \left(\varphi_k\varphi_{k'}\left(n''\right)\right), (kk')k''\right) \\
&= \left(n\ \varphi_k\left(n'\varphi_{k'}\left(n''\right)\right), k\left(k'k''\right)\right). \qquad (2.65) \\
(n,k) * \left[\left(n',k'\right) * \left(n'',k''\right)\right] &= (n,k) * \left(n'\varphi_{k'}\left(n''\right), k'k''\right) \\
&= \left(n\ \varphi_k\left(n'\varphi_{k'}\left(n''\right)\right), k\left(k'k''\right)\right). \qquad (2.66)
\end{aligned}$$

We claim that (e_N, e_K) is the identity.

$$(e_N, e_K) * (n, k) = (e_N \varphi_{e_K}(n), e_K k) \tag{2.67}$$
$$(n, k) * (e_N, e_K) = (n\varphi_k(e_N), k e_K) = (n, k). \tag{2.68}$$

We claim that $(n,k)^{-1} = \left(\varphi_{k^{-1}}\left(n^{-1}\right), k^{-1}\right)$.

$$\begin{aligned}(n,k) * \left(\varphi_{k^{-1}}\left(n^{-1}\right), k^{-1}\right) &= \left(n\varphi_k\left(\varphi_{k^{-1}}\left(n^{-1}\right)\right), kk^{-1}\right) \\ &= \left(nn^{-1}, kk^{-1}\right) = (e_N, e_K) \qquad (2.69)\\ \left(\varphi_{k^{-1}}\left(n^{-1}\right), k^{-1}\right) * (n,k) &= \left(\varphi_{k^{-1}}\left(n^{-1}\right)\varphi_{k^{-1}}(n), k^{-1}k\right) \\ &= \left(\varphi_{k^{-1}}\left(n^{-1}n\right), k^{-1}k\right) = (e_N, e_K). \qquad (2.70)\end{aligned}$$

Therefore, $N \rtimes_\varphi K$ is a group.

(b) Let's first verify that i and p are group homomorphisms:

$$\begin{aligned}i(n_1) * i(n_2) = (n_1, e_K) * (n_2, e_K) &= (n_1 \varphi_{e_K}(n_2), e_K e_K) \qquad (2.71)\\ &= (n_1 n_2, e_K) = i(n_1 n_2). \qquad (2.72)\\ p\left((n,k) * \left(n', k'\right)\right) = p\left(n\ \varphi_k\left(n'\right), kk'\right) &= kk' = p(n,k)\, p\left(n', k'\right). \qquad (2.73)\end{aligned}$$

Clearly i is injective and p is surjective. Also,

$$\operatorname{im} i = N \times \{e_K\} = \operatorname{Ker} p. \tag{2.74}$$

Hence, the sequence $e \to N \to N \rtimes_\varphi K \to K \to e$ is exact.

(c) If φ is the trivial homomorphism, then $\varphi_k = \mathrm{id}_N$ for every k. Then

$$(n,k) * \left(n', k'\right) = \left(n\ \varphi_k\left(n'\right), kk'\right) = \left(n\ \mathrm{id}_N\left(n'\right), kk'\right) = \left(nn', kk'\right), \tag{2.75}$$

which is exactly the same as the group operation on the direct product $N \times K$.

(d) We define the map

$$\begin{aligned}f : N \rtimes_\varphi K &\to G \\ (n,k) &\mapsto nk.\end{aligned} \tag{2.76}$$

Let's first check that it's a homomorphism:

$$\begin{aligned}f\left((n,k) * \left(n', k'\right)\right) = f\left(\left(n\ \varphi_k\left(n'\right), kk'\right)\right) &= f\left(n\ kn'k^{-1}, kk'\right) \qquad (2.77)\\ &= nkn'k^{-1}\ kk' = nk\ n'k' \qquad (2.78)\\ &= f(n,k)\, f\left(n', k'\right). \qquad (2.79)\end{aligned}$$

It's injective because if $f(n,k) = e$, then

$$e = f(n,k) = nk \implies n = k^{-1} \in N \cap K = \{e\}, \tag{2.80}$$

so $\operatorname{Ker} f = \{(e,e)\}$. Also, f is surjective since $G = NK$, so every element $g \in G$ can be written as $g = nk$ for some $n \in N$ and $k \in K$. Therefore, f is an isomorphism.

■

§3 Problems on Groups from Lothar Göttsche's Notes

Exercise 3.1. Find a subgroup H of S_3 of order 3, i.e. with $|H| = 3$.

Solution. Consider

$$H = \langle (1\ 2\ 3) \rangle = \{e, (1\ 2\ 3), (1\ 3\ 2)\}. \tag{3.1}$$

Then $|H| = 3$, and H is a subgroup of S_3. In fact, H is isomorphic to $\mathbb{Z}_3$. ■

Exercise 3.2. Let G be a group. The center $C(G)$ of G is

$$C(G) := \{g \in G \mid ga = ag,\ \forall a \in G\}. \tag{3.2}$$

(a) Show $C(G)$ is a subgroup of G.

(b) Show $C(G)$ is commutative.

(c) What is the center of S_3?

Solution. (a) First, $C(G) \neq \varnothing$, since $e \in C(G)$. Now, given any $x, y \in C(G)$, we'll show that $xy \in C(G)$ and $x^{-1} \in C(G)$. First notice that

$$(xy)\, a = x\,(ya) = x\,(ay) = (xa)\, y = (ax)\, y = a\,(xy), \tag{3.3}$$

for every $a \in G$; and hence, $xy \in C(G)$. Now,

$$xa = ax \implies x^{-1}(xa)\, x^{-1} = x^{-1}(ax)\, x^{-1} \implies ax^{-1} = x^{-1}a, \tag{3.4}$$

for every $a \in G$; and hence, $x^{-1} \in C(G)$. Therefore, $C(G) \leq G$.

(b) The elements of $C(G)$ commute with **every** element of G. So, in particular, they commute among themselves. Therefore, $C(G)$ is abelian.

(c) We claim that $C(S_3) = \{e\}$.

$$(1\ 2)(1\ 2\ 3) = (2\ 3), \quad (1\ 2\ 3)(1\ 2) = (1\ 3). \tag{3.5}$$

Hence, neither (1 2), nor (1 2 3) belong in the center. Since center is a subgroup, $(1\ 2\ 3)^{-1} = (1\ 3\ 2)$ is also not in the center. We can rule out (2 3) and (1 3) similarly:

$$(2\ 3)(1\ 2\ 3) = (1\ 3), \quad (1\ 2\ 3)(2\ 3) = (1\ 2); \tag{3.6}$$

$$(1\ 3)(1\ 2\ 3) = (1\ 2), \quad (1\ 2\ 3)(1\ 3) = (2\ 3). \tag{3.7}$$

Therefore, $C(S_3) = \{e\}$.

■

Exercise 3.3. Prove that every subgroup of a cyclic group is cyclic.

Solution. Let $G = \langle a \rangle$ be a cyclic group and $H \leq G$. Then every element of H is of the form a^i for some $i \in \mathbb{Z}$. Let

$$m = \min \left\{ i \in \mathbb{Z}_{>0} \mid a^i \in H \right\}. \tag{3.8}$$

We claim that $H = \langle a^m \rangle$. Since $a^m \in H$, it's clear that $\langle a^m \rangle \subset H$. Assume for the sake of contradiction that there exists some $b \in H$ and $b \notin \langle a^m \rangle$. Since $b \in G = \langle a \rangle$,

$$b = a^n, \quad \text{for some } n \in \mathbb{Z}. \tag{3.9}$$

Now, by performing division algorithm on n, we can write

$$n = mq + r, \quad \text{for } 0 \leq r < m. \tag{3.10}$$

Since $b \notin \langle a^m \rangle$, $r > 0$. Now, since $b = a^{mq+r}, a^m \in H$, $a^r \in H$. But m is the minimal positive integer such that $a^m \in H$, and now we have found an even smaller positive integer. Contradiction! ■

Exercise 3.4. Let G be a finite group. Let $K \subset H$ and $H \subset G$ be subgroups. Show that

$$[G : K] = [G : H][H : K]. \tag{3.11}$$

Solution. We shall use the fact that for finite groups, $[G : H] = \frac{|G|}{|H|}$.

$$[G : H][H : K] = \frac{|G|}{|H|} \cdot \frac{|H|}{|K|} = \frac{|G|}{|K|} = [G : K]. \tag{3.12}$$

■

Exercise 3.5. Let G be a group and H, K normal subgroups. Then $H \cap K$ is a normal subgroup of G.

Solution. It's easy to see that $N := H \cap K$ is a subgroup of G. Indeed, $N \neq \varnothing$ as $e \in N$. For $x, y \in N$,

$$x, y \in H \text{ implies } xy^{-1} \in H; \text{ and } x, y \in K \text{ implies } xy^{-1} \in K. \tag{3.13}$$

Therefore, $xy^{-1} \in H \cap K = N$. Hence, $N \leq G$. Now, for any $g \in G$,

$$N \subseteq H \implies gNg^{-1} \subseteq gHg^{-1} = H, \quad \text{and } N \subseteq K \implies gNg^{-1} \subseteq gKg^{-1} = K. \tag{3.14}$$

Hence, $gNg^{-1} \subseteq H \cap K = N$, proving that $N \trianglelefteq G$. ■

Exercise 3.6. Let G be a group and H a subgroup of index 2. Show H is a normal subgroup of G.

Solution. We shall prove a more general statement:

Suppose G is a finite group, and let p be the smallest prime divisor of $|G|$. If H is a subgroup with $[G : H] = p$, then $H \trianglelefteq G$.

Consider the action of G on the left coset space of H, by left multiplication:

$$g \cdot (aH) := (ga) H. \tag{3.15}$$

This gives a group homomorphism

$$f : G \to \mathrm{Sym}\,(G/H) \cong S_p. \tag{3.16}$$

Then $\mathrm{Ker}\, f \trianglelefteq G$. Furthermore, if $k \in \mathrm{Ker}\, f$, then for all $aH \in G/H$,

$$k \cdot (aH) := (ka) H = aH. \tag{3.17}$$

In particular, choosing $a = e$ gives us $kH = H$, i.e. $k \in H$. Hence,

$$\mathrm{Ker}\, f \leq H \leq G. \tag{3.18}$$

By the first isomorphism theorem, $\mathrm{im}\, f \cong \frac{G}{\mathrm{Ker}\, f}$, and hence $|\mathrm{im}\, f|$ divides $|G|$. But since $\mathrm{im}\, f$ is (isomorphic to) a subgroup of S_p, it divides $|S_p| = p!$ as well. Therefore,

$$|\mathrm{im}\, f| \,\Big|\, \gcd\left(|G|, p!\right) = p, \tag{3.19}$$

since p is the smallest prime factor of $|G|$. Therefore, $|\mathrm{im}\, f| = |G/\mathrm{Ker}\, f| = [G : \mathrm{Ker}\, f]$ divides p. From (3.18), we have

$$[G : \mathrm{Ker}\, f] = [G : H]\,[H : \mathrm{Ker}\, f] = p\,[H : \mathrm{Ker}\, f], \tag{3.20}$$

which divides p. Hence, we must have $[H : \mathrm{Ker}\, f] = 1$, i.e. $H = \mathrm{Ker}\, f$. Since kernels are always normal, $H \trianglelefteq G$. ■

Exercise 3.7. Let G be a group. Let G' be the subgroup of G generated by

$$U = \{xyx^{-1}y^{-1} \mid x, y \in G\}. \tag{3.21}$$

(a) Show G' is a normal subgroup of G.

(b) Show G/G' is abelian.

Solution. (a) We denote $[x, y] := xyx^{-1}y^{-1}$. If $f : G \to G$ is a homomorphism, then

$$f\left([x, y]\right) = f\left(xyx^{-1}y^{-1}\right) = f(x)\, f(y)\, f(x)^{-1}\, f(y)^{-1} = \left[f(x), f(y)\right]. \tag{3.22}$$

Therefore, $f(U) \subseteq U$. As a result,

$$f\left(G'\right) = f\left(\langle U \rangle\right) \subseteq \langle U \rangle = G'. \tag{3.23}$$

For any $g \in G$, $x \mapsto gxg^{-1}$ is a homomorphism, so $gG'g^{-1} \subseteq G'$ for any $g \in G$. Hence, $G' \trianglelefteq G$.

(b)

$$\begin{aligned} \left(xG'\right)\left(yG'\right) = \left(yG'\right)\left(xG'\right) &\iff (xy)\,G' = (yx)\,G' \\ &\iff (yx)^{-1}\,(xy) \in G' \\ &\iff x^{-1}y^{-1}xy \in G', \end{aligned} \tag{3.24}$$

which is true! Hence, G/G' is abelian.

■

Exercise 3.8. Let G be a group. Show that $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$.

Solution. First, we'll show that $\mathrm{Inn}(G) \leq \mathrm{Aut}(G)$. Let $\tau_g, \tau_h \in \mathrm{Inn}(G)$, where

$$\tau_g\,(x) = gxg^{-1}, \quad \tau_h\,(x) = hxh^{-1}. \tag{3.25}$$

Then

$$(\tau_g \circ \tau_h)\,(x) = \tau_g\left(hxh^{-1}\right) = g\left(hxh^{-1}\right)g^{-1} = (gh)\,x\,(gh)^{-1} = \tau_{gh}(x). \tag{3.26}$$

Hence, $\tau_g \circ \tau_h = \tau_{gh} \in \mathrm{Inn}(G)$. Furthermore, we claim that $\tau_g^{-1} = \tau_{g^{-1}}$.

$$\tau_g \circ \tau_{g^{-1}} = \tau_{gg^{-1}} = \tau_e = \mathrm{id}_G, \quad \tau_{g^{-1}} \circ \tau_g = \tau_{g^{-1}g} = \tau_e = \mathrm{id}_G\,. \tag{3.27}$$

Hence, $\tau_g^{-1} = \tau_{g^{-1}} \in \mathrm{Inn}(G)$.

Now we show $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$. Let $\varphi \in \mathrm{Aut}(G)$, and $\tau_g \in \mathrm{Inn}(G)$.

$$\left(\varphi \circ \tau_g \varphi^{-1}\right)(x) = \varphi\left(g \cdot \varphi^{-1}(x) \cdot g^{-1}\right) = \varphi(g) \cdot x \cdot \varphi(g)^{-1} = \tau_{\varphi(g)}(x). \tag{3.28}$$

Hence, $\varphi \circ \tau_g \varphi^{-1} = \tau_{\varphi(g)} \in \mathrm{Inn}(G)$, proving that $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$. ■

Exercise 3.9. Give an example of a group G and subgroups $K \subset H \subset G$, such that K is a normal subgroup of H and H is a normal subgroup of G, but K is not a normal subgroup of G.

Solution. Consider

$$G = D_4 = \left\{e, r, r^2, r^3, s, sr, sr^2, sr^3\right\}, \tag{3.29}$$

$$H = \left\{e, r^2, s, sr^2\right\}, \tag{3.30}$$

$$K = \{e, s\}\,. \tag{3.31}$$

Then

$$[G : H] = [H : K] = 2, \tag{3.32}$$

so $K \trianglelefteq H \trianglelefteq G$. But $K \not\trianglelefteq G$, since

$$rsr^{-1} = r\left(sr^{-1}\right) = r\left(rs\right) = r^2 s \notin K. \tag{3.33}$$

Therefore, K is not a normal subgroup of G. ■

Exercise 3.10. Let H, K be normal subgroups of a group G with $H \cap K = \{1\}$. Show that

$$hk = kh \quad \text{for all } h \in H,\ k \in K. \tag{3.34}$$

Solution. Consider the element $hkh^{-1}k^{-1}$.

$$hkh^{-1}k^{-1} = \left(hkh^{-1}\right)k^{-1} \in K, \tag{3.35}$$

by the normality of K. Again,

$$hkh^{-1}k^{-1} = h\left(kh^{-1}k^{-1}\right) \in H, \tag{3.36}$$

by the normality of H. Therefore, $hkh^{-1}k^{-1} \in H \cap K = \{e\}$, so

$$hkh^{-1}k^{-1} = e \implies hk = kh. \tag{3.37}$$

■

Exercise 3.11. Let

$$\sigma := \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 3 & 4 & 5 & 1 & 6 & 7 & 9 & 8 \end{pmatrix} \in S_9. \tag{3.38}$$

Find the cycle decomposition of σ, find the cycle type of σ, and write an element of S_9 which is conjugated to σ.

Solution. One can check that

$$1 \to 2 \to 3 \to 4 \to 5 \to 1, \quad 8 \to 9 \to 8, \tag{3.39}$$

and $6, 7$ are fixed points. Therefore,

$$\sigma = (1\ 2\ 3\ 4\ 5)\,(6)\,(7)\,(8\ 9)\,. \tag{3.40}$$

So its cycle-type is $(5, 2, 1, 1)$. Any element of this cycle-type will be conjugate to σ, i.e. any

$$\tau = (a_1\ a_2\ a_3\ a_4\ a_5)\,(a_6)\,(a_7)\,(a_8\ a_9) \tag{3.41}$$

is conjugate to σ. ■

Exercise 3.12. In the following, verify if the maps $\varphi : G \to G$ are homomorphisms of groups. If they are, determine their kernel.

(a) $G = (\mathbb{R} \setminus \{0\},\ \cdot\ )$, the nonzero real numbers with multiplication, $\varphi(x) = x^2$.

(b) $G = (\mathbb{R}, +)$, the real numbers with addition. $\varphi(x) = x + 1$.

(c) G is an abelian group, $\varphi(x) = x^3$.

Solution. (a) Yes, $\varphi : \mathbb{R}^\times \to \mathbb{R}^\times$, $x \mapsto x^2$ is a homomorphism, since

$$\varphi(xy) = (xy)^2 = x^2y^2 = \varphi(x)\,\varphi(y). \tag{3.42}$$

Its kernel is

$$\operatorname{Ker}\varphi = \left\{x \in \mathbb{R}^\times \mid \varphi(x) = 1\right\} = \left\{x \in \mathbb{R}^\times \mid x^2 = 1\right\} = \{1, -1\}. \tag{3.43}$$

(b) $\varphi : (\mathbb{R}, +) \to (\mathbb{R}, +)$, $x \mapsto x + 1$ is not a homomorphism, since $\varphi(0) \neq 0$.

(c) If G is abelian, $\varphi : G \to G$, $x \mapsto x^3$ is a homomorphism, since

$$\varphi(xy) = (xy)^3 = (xy)(xy)(xy) = (xxx)(yyy) = x^3y^3 = \varphi(x)\,\varphi(y). \tag{3.44}$$

Its kernel is

$$\operatorname{Ker}\varphi = \left\{x \in G \mid x^3 = e\right\} = \{x \in G \mid \operatorname{ord}(x) = 1 \text{ or } 3\}. \tag{3.45}$$

■

Exercise 3.13. Let G be a group, let $\varphi : G \to G$ be an automorphism, and let $Z(G)$ be the center of G. Show

$$\varphi(Z(G)) \subseteq Z(G). \tag{3.46}$$

Solution. For $z \in Z(G)$, z commutes with every $x \in G$. Hence,

$$zx = xz \implies \varphi(zx) = \varphi(xz)\varphi(z)\varphi(x) = \varphi(x)\varphi(z). \tag{3.47}$$

Since φ is an automorphism, given any $g \in G$, we can find $x \in G$ such that $\varphi(x) = g$. Hence, for any $g \in G$,

$$\varphi(z)\ g = g\ \varphi(z), \tag{3.48}$$

i.e. $\varphi(z) \in Z(G)$, proving that $\varphi(Z(G)) \subseteq Z(G)$. ■

In fact, $\varphi(Z(G)) = Z(G)$. The other containment can be shown using the fact that φ^{-1} is also an automorphism, so $\varphi^{-1}(Z(G)) \subseteq Z(G)$. As a result,

$$Z(G) = \varphi\left(\varphi^{-1}(Z(G))\right) \subseteq \varphi(Z(G)). \tag{3.49}$$

Exercise 3.14. Let G be a cyclic group of prime order. Determine $\operatorname{Aut}(G)$.

Solution. For $G = \mathbb{Z}_p$, we claim that $\mathrm{Aut}(G) \cong \mathbb{Z}_p^\times$. Define

$$\begin{aligned} \varphi : \mathrm{Aut}\left(\mathbb{Z}_p\right) &\to \mathbb{Z}_p^\times \\ f &\mapsto f(1). \end{aligned} \tag{3.50}$$

This is a homomorphism since

$$\varphi\left(f \circ g\right) = \left(f \circ g\right)(1) = f\left(g(1)\right) = f\left(g(1) \cdot 1\right) = g(1)f(1) = f(1)g(1) = \varphi(f)\varphi(g). \tag{3.51}$$

It's injective because if $f \in \operatorname{Ker}\varphi$, then $f\left(1\right) = 1$, and then $f\left(a\right) = a$ for each $a \in \mathbb{Z}_p$, i.e. $f = \mathrm{id}$.

Furthermore, φ is surjective, because given any $b \in \mathbb{Z}_p^\times$, we can define an automorphism $f\left(a\right) = ab$, for which $f\left(1\right) = b$. Hence, φ is an isomorphism. ■

Exercise 3.15. If $G/Z(G)$ is cyclic, then G is abelian.

Solution. For simplicity of notations, we'll denote $N := Z(G)$. Since G/N is cyclic,

$$G/N = \langle xN \rangle = \left\{N, xN, x^2N, \ldots, x^nN\right\}. \tag{3.52}$$

Let's take any two element $g, h \in G$. Since the cosets in G/N partition G, we must have that $g = x^i z_1$ and $h = x^j z_2$ for some $z_1, z_2 \in N = Z(G)$. Then

$$\begin{aligned} gh &= \left(x^i z_1\right)\left(x^j z_2\right) = x^i\left(z_1 x^j\right) z_2 && (3.53)\\ &= x^i\left(x^j z_1\right) z_2 = x^j x^i\left(z_1 z_2\right) && (3.54)\\ &= x^j\left[\left(x^i z_1\right) z_2\right] = x^j\left[z_2\left(x^i z_1\right)\right] && (3.55)\\ &= \left(x^j z_2\right)\left(x^i z_1\right) = hg. && (3.56) \end{aligned}$$

Hence, G is abelian. ■

Exercise 3.16. An r-cycle is even if and only if r is odd.

Solution.

$$\left(a_1\ a_2\ \cdots a_r\right) = \left(a_1\ a_2\right)\left(a_2\ a_3\right)\cdots\left(a_{r-1}\ a_r\right), \tag{3.57}$$

so a r-cycle is a product of $r-1$ transitions. So it is even if and only if $r-1$ is even, i.e. r is odd. ■

Exercise 3.17. Let G be a group of order 30.

(a) Show a 3-Sylow or a 5-Sylow subgroup is normal in G.

(b) Show every 3-Sylow and every 5-Sylow subgroup is normal in G.

(c) Show G has a normal subgroup of order 15.

Solution. We prove some preliminary result first.

Claim 3.1: If $N \trianglelefteq G$, and $H \trianglelefteq N$ is a p-Sylow subgroup of N, then $H \trianglelefteq G$.

Proof. Since $H \trianglelefteq N$, $n_p(N) = 1$. Then for any $g \in G$,

$$gHg^{-1} \subseteq gNg^{-1} = N. \tag{3.58}$$

Hence, gHg^{-1} is a p-Sylow subgroup of N, since $\left|gHg^{-1}\right| = |H|$. But since $n_p(N) = 1$, we must have that $gHg^{-1} = H$, for any $g \in G$. As a result, $H \trianglelefteq G$. □

(a) Assume for the sake of contradiction that neither a 3-Sylow subgroup nor a 5-Sylow subgroup is normal. Then $n_3, n_5 > 1$.

Since $n_5 \mid 6$ and $n_5 \equiv 1 \pmod 5$, then we must have $n_5 = 6$. Therefore, there are $6 \times 4 = 24$ elements of order 5 (since the 5-Sylow subgroups intersect trivially, because the non-identity elements generate the whole subgroup since it's a cyclic group). Hence,

$$2n_3 = (\text{number of elements of order } 3) \leq 30 - 24 - 1 = 5, \tag{3.59}$$

implying that $n_3 \leq 2$. Since $n_3 \equiv 1 \pmod 3$, we must have $n_3 = 1$.

(b) We have shown earlier that $n_3 = 1$. Now we'll show that $n_5 = 1$. Let H_3 be the 3-Sylow subgroup and H_5 be a 5-Sylow subgroup. Consider $H = H_3H_5 \leq G$. Then

$$|H| = \frac{|H_3|\,|H_5|}{|H_3 \cap H_5|} = 15, \tag{3.60}$$

and from the proof of Exercise 3.24, we conclude that H is cyclic. Since $[G : H] = 2$, H is normal in G. Furthermore, since H is cyclic, it has a unique 5-Sylow subgroup, which is H_5. Therefore, by Claim 3.1, $H_5 \trianglelefteq G$, i.e. $n_5(G) = 1$.

(c) As before, we consider the subgroup $H = H_3H_5 \leq G$, which has order 15, i.e. its index is 2. Therefore, it's normal.

■

Exercise 3.18. If G is a group of order 385, show that its 11-Sylow subgroup is normal and its 7-Sylow subgroup is in the center of G.

Solution. $|G| = 385 = 5 \cdot 7 \cdot 11$. $n_{11} \mid 35$, so

$$n_{11} \in \{1, 5, 7, 35\}, \tag{3.61}$$

and $n_{11} \equiv 1 \pmod{11}$ forces that we must have $n_{11} = 1$. Hence, the 11-Sylow subgroup is normal. Now, $n_7 \mid 55$ gives

$$n_7 \in \{1, 5, 11, 55\}, \tag{3.62}$$

and $n_7 \equiv 1 \pmod 7$ forces that we must have $n_7 = 1$. Hence, the 7-Sylow subgroup H is normal.

Claim 3.2: If $H \trianglelefteq G$, then $C_G(H) \trianglelefteq G$ and $G/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$.

Proof. Consider the action of G on H by conjugation. This gives rise to a group homomorphism

$$\begin{aligned} \Psi : G &\to \mathrm{Aut}(H) \\ g &\mapsto \left(h \mapsto ghg^{-1}\right). \end{aligned} \tag{3.63}$$

Then

$$\begin{aligned} \mathrm{Ker}\,\Psi &= \left\{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\right\} && (3.64) \\ &= \{g \in G \mid gh = hg \text{ for all } h \in H\} = C_G(H). && (3.65) \end{aligned}$$

Hence, $C_G(H) \trianglelefteq G$ and by the First isomorphism theorem, $G/C_G(H)$ is isomorphic to $\mathrm{im}\,\Psi \leq \mathrm{Aut}(H)$. □

Now, by Claim 3.2, $G/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$. Since $H \cong \mathbb{Z}_7$, by Exercise 3.14, $|\mathrm{Aut}(H)| = 6$. Therefore,

$$|G/C_G(H)| \,\Big|\, |\mathrm{Aut}(H)| = 6. \tag{3.66}$$

But $|G|$ is coprime with 6, and hence so is $|G/C_G(H)| = \frac{|G|}{|C_G(H)|}$. Therefore,

$$|G/C_G(H)| = 1 \implies G = C_G(H). \tag{3.67}$$

So every element of G commutes with every element of H, proving that $H \leq Z(G)$. ■

Exercise 3.19. Let G be a group of order 231. Show the 11-Sylow subgroup is in the center of G.

Solution. $|G| = 231 = 3 \cdot 7 \cdot 11$. Now, $n_{11} \mid 21$ gives

$$n_7 \in \{1, 3, 7, 21\}, \tag{3.68}$$

and $n_{11} \equiv 1 \pmod{11}$ forces that we must have $n_{11} = 1$. Hence, the 11-Sylow subgroup H is normal. Now, by Claim 3.2, $G/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$. Since $H \cong \mathbb{Z}_{11}$, by Exercise 3.14, $|\mathrm{Aut}(H)| = 10$. Therefore,

$$|G/C_G(H)| \,\Big|\, |\mathrm{Aut}(H)| = 10. \tag{3.69}$$

But $|G|$ is coprime with 10, and hence so is $|G/C_G(H)| = \frac{|G|}{|C_G(H)|}$. Therefore,

$$|G/C_G(H)| = 1 \implies G = C_G(H). \tag{3.70}$$

So every element of G commutes with every element of H, proving that $H \leq Z(G)$. ■

Exercise 3.20. If G is a group of order p^2q with p and q primes, show that G has a nontrivial normal subgroup.

Solution. We'll first consider the case $p^2q = 12$. Assume the contrary. Note that $12 = 2^2 \times 3$. Since $n_3 \mid 4$ and $n_3 > 1$ (since we assumed the group is simple) and $n_3 \equiv 1 \pmod 3$, we must have that $n_3 = 4$. This gives

$$\text{(number of elements of order 3)} = n_3(3-1) = 4 \cdot 2 = 8. \tag{3.71}$$

The 2-Sylow subgroups can't intersect any of the 3-Sylow subgroups. The identity element together with the other 3 elements form one 2-Sylow subgroup. Therefore, we don't have any room for any other 2-Sylow subgroups, implying $n_2 = 1$.

Now, we consider the general case: Assume for the sake of contradiction that ther exists a simple group G with $|G| = p^2q$. Note that

$$p + 1 \leq n_p \leq q. \tag{3.72}$$

The first bound follows from the fact that $n_p \neq 1$, and the second bound is given by $n_p \mid q$. Now, note that $n_q \mid p^2$ and $n_q > q > p$, so we must have $n_q = p^2$. This gives us

$$p^2 \equiv 1 \pmod q \implies q \mid (p-1)(p+1). \tag{3.73}$$

Since $q > p$, the scenario $q \mid p-1$ is impossible, so we must have $q \mid p+1$. But that way $q \leq p+1$. But we already know $q \geq p+1$. Therefore, $q = p+1$. This can only happen if $p = 2$, $q = 3$, i.e. $G = 12$, which has already been done! ■

Exercise 3.21. How many elements of order 5 are contained in a group of order 20?

Solution. Since $n_5 \mid 4$ and $n_5 \equiv 1 \pmod 5$, we must have $n_5 = 1$. Therefore, there exists a unique 5-Sylow subgroup. Hence, the number of elements of order 5 is 4. ■

Exercise 3.22. Prove: No group of order pq, with p and q prime numbers, is simple.

Solution. Assume for the sake of contradiction that ther exists a simple group G with $|G| = pq$. WLOG, $p < q$. Then $n_q \mid p$. But $n_q = p$ is not possible since $n_q > 1$ along with

$$n_q \equiv 1 \pmod q \tag{3.74}$$

would imply

$$n_q \geq q + 1 > p. \tag{3.75}$$

Hence, $n_q = 1$, and hence the q-Sylow subgroup is normal. ■

Exercise 3.23. Let G be a group of order $p^m r$. Show G contains a subgroup of order p^e for every $e \leq m$.

Solution. We proceed by induction on e. The base case $e = 0$ is trivial. Suppose H is a subgroup of order $p^e < p^m$, we shall construct a subgroup of order p^{e+1}.

Claim 3.3 (p-group lemma)**:** If a p-group (group of order p^n) acts on a set S, then if N is number of fixed points (i.e. singleton orbits) of this action,

$$N \equiv |S| \pmod{p}. \tag{3.76}$$

Proof. By Orbit-Stabilizer theorem, the possible orbit sizes are $1, p, p^2, \ldots, p^n$. Then

$$|S| = \sum_{\text{orbits}} |\mathrm{Orb}(x)| \equiv \sum_{\text{singleton orbits}} |\mathrm{Orb}(x)| = N \pmod{p}. \tag{3.77}$$

□

Claim 3.4 (Normalizer lemma)**:** If H is a p-subgroup of G, then

$$[N_G(H) : H] \equiv [G : H] \pmod{p}. \tag{3.78}$$

Proof. Let H act on the left cosets G/H by left multiplication

$$h \cdot (gH) = (hg)\, H. \tag{3.79}$$

gH is a fixed-point of this action if and only if

$$\begin{aligned} h \cdot (gH) = gH \text{ for all } h \in H &\iff \left(g^{-1}hg\right) H = H \text{ for all } h \in H && (3.80)\\ &\iff g^{-1}hg \in H \text{ for all } h \in H && (3.81)\\ &\iff g \in N_G(H). && (3.82) \end{aligned}$$

Therefore, the fixed points are precisely the cosets in $N_G(H)/H$. As a result, the number of fixed points is $[N_G(H) : H]$, and hence, by p-group lemma, $[N_G(H) : H] \equiv [G : H] \pmod{p}$. □

Now, since $H \trianglelefteq N_G(H)$, consider the quotient group $N_G(H)/H$. Since $p \mid [G : H] = p^{m-e} r$, by Normalizer lemma, $p \mid |N_G(H)/H|$. Therefore, by Cauchy's theorem, we can find an element of order p in $N_G(H)/H$.

Say $aH \in N_G(H)/H$ is an element of order p. Then $a^p \in H$, since it's of order p. Now, define

$$H' = \bigcup_{i=0}^{p-1} a^i H. \tag{3.83}$$

Then $H' \leq G$, because if we take $a^i h, a^j h_\prime \in H'$ for $h, h_\prime \in H$,

$$\left(a^i h\right)\left(a^j h_\prime\right)^{-1} = a^i h\ h_\prime^{-1} a^{-j} = a^i\ a^{-j}\ h' \in H' \tag{3.84}$$

for some $h' \in H$, since $a^{-j} \in N_G(H)$. So $H' \leq G$. Finally,

$$\left|H'\right| = \sum_{i=0}^{p-1} \left|a^i H\right| = \sum_{i=0}^{p-1} |H| = p\,|H| = p^{e+1}. \tag{3.85}$$

■

Exercise 3.24. Classify the groups of order 33.

Solution. We shall prove a general statement:

If G is a group of order pq and $p < q$ primes with $q \not\equiv 1 \pmod{p}$, then $G \cong \mathbb{Z}_{pq}$.

By the proof of Exercise 3.22, $n_q = 1$. $n_p \mid q$ and $q \not\equiv 1 \pmod{p}$ implies $n_p = 1$.

Therefore, $H_1 = \langle a \rangle, H_2 = \langle b \rangle$, the p-Sylow subgroup and the q-Sylow subgroup, respectively, are normal in G. Their intersection is trivial, since their orders are coprime. Therefore, a and b commute, since

$$aba^{-1}b^{-1} = \left(aba^{-1}\right)b^{-1} = a\left(ba^{-1}b^{-1}\right) \in H_1 \cap H_2 = \{e\}. \tag{3.86}$$

Now, a and b commute, their orders are coprime, so $\operatorname{ord}(ab) = pq$. Indeed,

$$(ab)^n = a^n b^n = e \iff a^n = b^{-n} \in H_1 \cap H_2 = \{e\} \tag{3.87}$$

$$\iff a^n = b^n = e \tag{3.88}$$

$$\iff p \mid n \text{ and } q \mid n. \tag{3.89}$$

So $n = pq$ is the smallest value to have $(ab)^n = e$. Now since $\operatorname{ord}(ab) = pq$, G is cyclic!

Now, for $|G| = 33 = 3 \times 11$, and $11 \not\equiv 1 \pmod{3}$, so any group of order 33 is cyclic. ■